# IMEN266 Operations Research II

## Concept & Proof Companion — Chapter 4 (Markov Chains)

[FULL version — all blanks filled]

Fall 2026

## How to use this document

This companion *replaces your class notes*: every definition and theorem from the Ch. 4 slides is written out here in full, so you can watch and think in class instead of transcribing. It does **not** replace *doing*: for each core proof, (1) work through the GAP-FILL version with pen and paper, (2) check against the FULL version, (3) answer the Checkpoints, and (4) if stuck — or done — use an AI study card. Exams are closed-book; these eight proofs and their checkpoints are exactly the kind of reasoning the exams reward. Items marked **(midterm-style)** mirror questions from past midterms: the subsampling proof (Proof 8), a path-factorisation true/false (Proof 1), $P^6$ from a given $P^3$ (Proof 2), a long-run cost (Proof 6), and PageRank with a dead end (Proof 7).

# 1 Concept digest

## 1.1 Stochastic processes

A **stochastic process** is an indexed family of random variables $\{X_t,\ t \in \mathcal{I}\}$: $X_t$ is a random variable for each fixed $t$, and $\mathcal{I}$ is the **index set** (usually time). Both the index set and the state space can be discrete or continuous:

| | index discrete | index continuous |
|---|---|---|
| state discrete | **Ch. 4: DTMC** (inventory at day start) | Ch. 5–7: Poisson process, CTMC, renewal |
| state continuous | time series (hourly temperature) | Brownian motion (financial engineering) |

*(verified in notebook: CP3 warm-up)*

## 1.2 Markov chains

$\{X_n,\ n \in \mathbb{Z}_+\}$ with countable state space $\mathcal{S}$ is a **(discrete-time) Markov chain** if, for all $n$ and all states,

$$P(X_{n+1} = j \mid X_n = i,\ X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i) \qquad \text{(Markov property)}$$

and, in this course, the right-hand side does not depend on $n$ (**time-homogeneity**): $P(X_{n+1} = j \mid X_n = i) = P(X_1 = j \mid X_0 = i) =: P_{ij}$. The **transition probability matrix** $P = (P_{ij})$ is *row-stochastic*: $P_{ij} \geq 0$ and $\sum_j P_{ij} = 1$ for every $i$. The **transition diagram** draws states as nodes and $P_{ij} > 0$ as arrows.

**Modeling recipe** (slide p. 7/52; every CP3 problem): (1) define $X_n$, the state at the $n$-th observation; (2) write $\mathcal{S}$; (3) verify Markov and time-homogeneity ("regardless of the history", "independent of the past"); (4) derive $P$ — and check that rows sum to 1; (5) draw the diagram. If the natural description has memory (age of a tool, day of a spell), *enlarge the state* until it is Markov (CP3 #6–#7). *(verified in notebook: CP3 #1–#7)*

## 1.3 Transient analysis

Initial distribution $a = [P(X_0 = 0), P(X_0 = 1), \dots]$; $p_n = [P(X_n = 0), P(X_n = 1), \dots]$ (row vectors). Then

$$p_1 = aP, \quad p_2 = p_1 P = aP^2, \quad \dots, \quad p_n = aP^n \qquad \text{(Proof 2)}.$$

The **$n$-step transition probabilities** $P_{ij}^{(n)} = P(X_n = j \mid X_0 = i)$ satisfy the **Chapman–Kolmogorov equations** $P_{ij}^{(m+n)} = \sum_k P_{ik}^{(m)} P_{kj}^{(n)}$, i.e. $P^{(n)} = P^n$ (matrix power). Path probabilities multiply along the path (Proof 1): $P(X_0 = i_0, \dots, X_n = i_n) = a_{i_0} P_{i_0 i_1} \cdots P_{i_{n-1} i_n}$. Observing the chain every $k$-th step gives a DTMC with matrix $P^k$ (Proof 8). *(verified in notebook: CP4 #1–#3)*

## 1.4 Classification of states (slides pp. 29–33/52)

State $j$ is **accessible** from $i$ ($i \to j$) if $P_{ij}^{(n)} > 0$ for some $n \geq 0$; $i$ and $j$ **communicate** ($i \leftrightarrow j$) if each is accessible from the other. Communication is an equivalence relation (Proof 3), so $\mathcal{S}$ splits into **communicating classes**; the chain is **irreducible** if there is only one class.

Let $f_i = P(\text{ever return to } i \mid X_0 = i)$. State $i$ is **recurrent** if $f_i = 1$ and **transient** if $f_i < 1$; equivalently (Proof 3) $i$ is recurrent iff $\sum_{n \geq 1} P_{ii}^{(n)} = \infty$. Recurrence and transience are **class properties**. A recurrent state is **positive recurrent** if the mean return time $m_i = E[\text{return time} \mid X_0 = i]$ is finite (**null recurrent** otherwise); in a *finite* chain every recurrent state is positive recurrent, and a class is recurrent iff it is *closed* (no arrow leaves it). A state with $P_{ii} = 1$ is **absorbing**.

The **period** of $i$ is $d(i) = \gcd\{n \geq 1 : P_{ii}^{(n)} > 0\}$; $i$ is **aperiodic** if $d(i) = 1$ (a self-loop $P_{ii} > 0$ suffices). Period is also a class property. **Ergodic** = irreducible + positive recurrent + aperiodic (slide p. 34/52: the "desired" chains). *(verified in notebook: CP5 warm-up, `DTMC.summary()`)*

## 1.5 Limiting behavior (slide p. 36/52)

**Theorem.** If the chain is irreducible and positive recurrent, the **stationary equations**

$$\pi = \pi P, \qquad \sum_{i \in \mathcal{S}} \pi_i = 1$$

have a unique solution $\pi$; $\pi_i$ is the long-run fraction of time spent in $i$, and $\pi_i = 1/m_i$ (Proof 6). If the chain is also aperiodic (ergodic), then $\lim_{n \to \infty} P(X_n = i) = \pi_i$ for every initial distribution, i.e. every row of $P^n$ converges to $\pi$ (Proof 4 for two states).

*Reading the theorem.* Periodic chains (the 3-cycle) still have a unique $\pi$ but no limit; reducible chains may have limits that depend on the starting state (slide p. 29/52, CP5 warm-up). One of the balance equations is always redundant — replace it by the normalisation (CP5 #2).

## 1.6 Cost and reward (slide p. 45/52)

If cost $C(i)$ is incurred every time the chain is observed in state $i$, the long-run average cost per observation is

$$\lim_{n \to \infty} \frac{1}{n} \sum_{k=1}^{n} C(X_k) = \sum_{i \in \mathcal{S}} C(i)\, \pi_i \qquad \text{(Proof 6)}.$$

Examples: availability (CP5 #1), wet trips $\varphi = p\, \pi_0$ and $\tau(r) = c\varphi + dr$ (CP5 #3), mean tool age $\sum_i i\pi_i$ and mean lifetime $1/\pi_0$ (CP5 #4), lunch revenue (CP5 #5).

### 1.7 PageRank (handout)

A random surfer on $n$ web pages follows one of the $k$ outgoing links of the current page with probability $1/k$: a DTMC with matrix $P$; the rank of page $j$ is $\pi_j$. **Dead ends** (no outgoing links: zero row, mass leaks) and **spider traps** (absorbing pages: $\pi$ concentrates there) break ergodicity. **Taxation** with rate $1-\beta$ replaces $P$ by

$$G = \beta P + \frac{1-\beta}{n}\,\mathbf{1}\mathbf{1}^{\top},$$

a strictly positive stochastic matrix, so $\pi = \pi G$ has a unique solution and power iteration $p_{k+1} = p_k G$ converges at rate $\le \beta$ (Proof 7). Handout numbers: $\pi = (0.4, 0.4, 0.2)$ for the 3-page web; with the spider trap and 30 % tax, $\pi \approx (0.26, 0.19, 0.55)$. *(verified in notebook: PageRank practicum)*

### 1.8 Chain zoo (the class-problem models)

| model (CP3) | state $X_n$ | $\lvert\mathcal{S}\rvert$ | ergodic? | stationary distribution |
|---|---|---|---|---|
| machine (#1) | down/up | 2 | yes | $\pi_{\text{down}} = \frac{b}{a+b}$, $a = P_{\text{du}}$, $b = P_{\text{ud}}$ (Proof 4) |
| Pohang weather (#2) | S/C/R | 3 | yes | solve $3\times 3$: $(0.479, 0.311, 0.210)$ |
| Himart (#3) | Monday stock | 4 | yes | numeric; tail probabilities in column "5" |
| two routes (#4) | # free routes | 3 | yes | product of two 2-state chains |
| umbrellas (#5) | # at current place | $r+1$ | yes | $\pi_0 = \frac{1-p}{r+1-p}$, $\pi_i = \frac{1}{r+1-p}$ (Proof 5) |
| tool age (#6) | age | $K+1$ | yes | $\pi_i = \pi_0 \prod_{k<i}(1-p_k)$, $1/\pi_0 =$ mean lifetime |
| weather spells (#7) | (type, day of spell) | $2K$ | yes | numeric |
| Kim's lunch (CP5 #5) | restaurant | 3 | yes | $(10, 5, 4)/19$ |

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## 2 Core proofs

Notation: $\mathcal{S}$ finite unless stated; $a$ is the initial distribution; $P^{(n)}_{ij} = P(X_n = j \mid X_0 = i)$. Blanks are the steps *you* must supply in the gap-fill version.

### Proof 1 — Path probabilities: the Markov property in action

**Theorem.** For any states $i_0, \dots, i_n$,

$$P(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) = a_{i_0}\, P_{i_0 i_1} P_{i_1 i_2} \cdots P_{i_{n-1} i_n},$$

and consequently $P(X_1 = j) = \sum_i a_i P_{ij}$, i.e. $p_1 = aP$.

*Proof.* By the multiplication rule (Ch. 1–3, Companion §1.2) applied $n$ times,

$$P(X_0 = i_0, \dots, X_n = i_n) = P(X_0 = i_0) \prod_{k=1}^{n} P(X_k = i_k \mid X_{k-1} = i_{k-1}, \dots, X_0 = i_0).$$

By the Markov property each conditional probability equals $P(X_k = i_k \mid X_{k-1} = i_{k-1})$, and by time-homogeneity this is $P_{i_{k-1} i_k}$, independent of $k$. Multiplying gives the product. For the consequence, the events $\{X_0 = i\}$, $i \in \mathcal{S}$, partition $\Omega$, so by the law of total probability $P(X_1 = j) = \sum_i P(X_1 = j \mid X_0 = i) P(X_0 = i) = \sum_i a_i P_{ij}$. $\square$

*(verified in notebook: CP4 #3(a),(b):* $0.3 \cdot 0.25 \cdot 0.3 \cdot 0.1$ *matched the simulation)*

**Checkpoint** — answer before moving on.

- In CP4 #3(b), $P[X_3 = 3, X_2 = 2, X_1 = 1]$ has no $X_0$: which step of the proof is replaced by a sum, and over what?
- Write the same product for a chain that is Markov but *not* time-homogeneous (transition matrices $P^{(k)}$ at step $k$).
- Why does the factorisation fail for the weather-spell process if the state is only "sunny/rainy" (CP3 #7)?
- **(midterm-style)** True or false, with a reason: $P(X_6 = 2, X_3 = 1, X_1 = 0 \mid X_0 = 2) = P(X_6 = 2 \mid X_0 = 2)\, P(X_3 = 1 \mid X_0 = 2)\, P(X_1 = 0 \mid X_0 = 2)$. Write the correct factorisation.

**AI study card** — paste one of these into your AI assistant:

- `Quiz me one step at a time on why P(X0=i0,...,Xn=in) factorises for a Markov chain. Make me say exactly where the Markov property and where time-homogeneity are used.`
- `Invent a 3-state process that is NOT Markov, show me a path whose probability does not factorise, and make me compute the true probability.`

## Proof 2 — Chapman–Kolmogorov and $p_n = aP^n$

**Theorem.** $P_{ij}^{(m+n)} = \sum_{k\in\mathcal{S}} P_{ik}^{(m)} P_{kj}^{(n)}$ for all $m, n \geq 0$; hence $P^{(n)} = P^n$ and $p_n = aP^n$.

*Proof.* Condition on the state at time $m$ (law of total probability):

$$P_{ij}^{(m+n)} = P(X_{m+n} = j \mid X_0 = i) = \sum_k P(X_{m+n} = j \mid X_m = k, X_0 = i)\, P(X_m = k \mid X_0 = i).$$

By the Markov property the first factor does not depend on $X_0$, and by time-homogeneity it equals $P_{kj}^{(n)}$; the second factor is $P_{ik}^{(m)}$. The sum is the $(i,j)$ entry of the matrix product $P^{(m)}P^{(n)}$. Taking $m = 1$ and inducting on $n$: $P^{(n)} = P\,P^{(n-1)} = P^n$. Finally $P(X_n = j) = \sum_i P(X_0 = i)P_{ij}^{(n)} = (aP^n)_j$. ☐

*(verified in notebook: CP4 #1–#2:* $aP^n$ *vs. 200,000 simulated paths)*

**Checkpoint** — answer before moving on.

- Explain in words why $P(X_5 = 2 \mid X_1 = 3) = (P^4)_{32}$ (CP4 #3(c)) — which two properties are used?
- Compute $P^2$ for the Pohang weather chain by hand and confirm $(P^2)_{SR} = 0.21$.
- True or false: $P_{ik}^{(m)} P_{kj}^{(n)}$ is the probability of going $i \to k \to j$ in $m + n$ steps *via* $k$ *at time* $m$. What does summing over $k$ do?
- **(midterm-style)** For $P = \begin{pmatrix} 0.5 & 0 & 0.5 \\ 0 & 0 & 1 \\ 0.3 & 0.2 & 0.5 \end{pmatrix}$ Python reports $P^3 = \begin{pmatrix} 0.35 & 0.10 & 0.55 \\ 0.30 & 0.10 & 0.60 \\ 0.33 & 0.12 & 0.55 \end{pmatrix}$. Find $P(X_{10008} = 1 \mid X_{10002} = 2)$ *without* computing $P^6$ from scratch. (Which C–K product? Compare with $\pi_1$ from Proof 6's checkpoint.)

**AI study card** — paste one of these into your AI assistant:

- `Play a skeptical student: I will prove the Chapman-Kolmogorov equations; interrupt whenever I use a property without naming it.`
- `Give me a 3x3 transition matrix and ask me for P(X_4=j | X_1=i) two ways: by matrix power and by summing over paths. Check that my two answers agree.`

## Proof 3 — Recurrence: the $\sum_n P_{ii}^{(n)}$ criterion and class property

**Theorem.** (a) State $i$ is recurrent iff $\sum_{n\geq 1} P_{ii}^{(n)} = \infty$. (b) If $i$ is recurrent and $i \leftrightarrow j$, then $j$ is recurrent. (c) A finite irreducible chain is recurrent.

*Proof of (a).* Let $N = \sum_{n\geq 1} \mathbf{1}\{X_n = i\}$ be the number of returns to $i$. Each time the chain is in $i$ it returns again with probability $f_i$, independently of the past (Markov property), so $P(N \geq k \mid X_0 = i) = f_i^k$ and by the tail-sum formula (Ch. 1–3, Proof 3)

$$E[N \mid X_0 = i] = \sum_{k\geq 1} f_i^k = \frac{f_i}{1-f_i} \quad (f_i < 1), \qquad E[N \mid X_0 = i] = \infty \quad (f_i = 1).$$

On the other hand, by linearity of expectation $E[N \mid X_0 = i] = \sum_{n\geq 1} E[\mathbf{1}\{X_n = i\} \mid X_0 = i] = \sum_{n\geq 1} P_{ii}^{(n)}$. So the sum is infinite iff $f_i = 1$. □

*Proof of (b).* Since $i \leftrightarrow j$ there are $k, m$ with $P_{ij}^{(k)} > 0$ and $P_{ji}^{(m)} > 0$. One way to go $j \to j$ in $m + n + k$ steps is $j \to i$ in $m$, $i \to i$ in $n$, $i \to j$ in $k$, so

$$P_{jj}^{(m+n+k)} \;\geq\; P_{ji}^{(m)}\, P_{ii}^{(n)}\, P_{ij}^{(k)}.$$

Summing over $n$: $\sum_n P_{jj}^{(m+n+k)} \geq P_{ji}^{(m)} P_{ij}^{(k)} \sum_n P_{ii}^{(n)} = \infty$ by (a), so $j$ is recurrent by (a). □

*Proof of (c).* If all states were transient, each is visited only finitely often, so after some finite time the chain would be in *no* state — impossible. Hence some state is recurrent, and by (b) and irreducibility all states are. □

*Remark.* In a finite chain a class is recurrent iff it is closed; transient classes are those with an arrow leaving them (slide p. 29/52: $\{3, 5\}$ leaks into the absorbing state 4). *(verified in notebook: CP5 warm-up,* `summary()`*)*

**Checkpoint** — answer before moving on.

- Where did the proof of (a) use the Markov property? (Why are successive returns independent trials with the same success probability?)
- Symmetric random walk on $\mathbb{Z}$ is recurrent but its mean return time is infinite (null recurrent). Why can this not happen in a finite chain?
- Classify every state of the two matrices on slide p. 29/52 and give the period of each class.

**AI study card** — paste one of these into your AI assistant:

- `Ask me to prove that recurrence is a class property. Do not accept "obvious"; make me write the inequality between n-step probabilities.`
- `Give me five small transition matrices; for each I must state the classes, which are recurrent, and the period. Grade me and explain any mistake with a path argument.`

## Proof 4 — The two-state chain: $P^n$ in closed form and its limit

**Theorem.** Let $P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ with $0 < a+b < 2$ (states $0,1$; $a = P_{01}$, $b = P_{10}$). Then with $\lambda = 1-a-b$,

$$P_{00}^{(n)} = \frac{b}{a+b} + \frac{a}{a+b}\lambda^n, \qquad P_{10}^{(n)} = \frac{b}{a+b} - \frac{b}{a+b}\lambda^n, \qquad \text{so} \quad P^n \to \begin{pmatrix} \frac{b}{a+b} & \frac{a}{a+b} \\ \frac{b}{a+b} & \frac{a}{a+b} \end{pmatrix} = \begin{pmatrix} \pi \\ \pi \end{pmatrix}.$$

*Proof.* Let $x_n = P_{00}^{(n)}$. By Chapman–Kolmogorov (condition on the state at time $n$),

$$x_{n+1} = P_{00}^{(n)} P_{00} + P_{01}^{(n)} P_{10} = x_n(1-a) + (1-x_n)\, b = b + \lambda\, x_n.$$

The fixed point of $x \mapsto b + \lambda x$ is $x^* = \frac{b}{1-\lambda} = \frac{b}{a+b}$, and subtracting $x^* = b + \lambda x^*$ gives $x_{n+1} - x^* = \lambda(x_n - x^*)$, hence $x_n - x^* = \lambda^n(x_0 - x^*)$ with $x_0 = P_{00}^{(0)} = 1$. So $x_n = \frac{b}{a+b} + \lambda^n\big(1 - \frac{b}{a+b}\big) = \frac{b}{a+b} + \frac{a}{a+b}\lambda^n$. The same recursion with $x_0 = P_{10}^{(0)} = 0$ gives the second formula. Since $0 < a+b < 2$ means $|\lambda| < 1$, $\lambda^n \to 0$ and both rows converge to $\pi = \big(\frac{b}{a+b}, \frac{a}{a+b}\big)$, which indeed solves $\pi = \pi P$. □

*Machine (CP5 #1).* down$=0$, up$=1$: $a = 0.97$, $b = 0.02$, $\pi_{\text{down}} = \frac{0.02}{0.99} \approx 0.0202$, $\lambda = 0.01$ — the chain forgets its start almost immediately. *(verified in notebook: CP5 #1 experiment: speed of forgetting $= |\lambda|$)*

**Checkpoint** — answer before moving on.

- What happens when $a+b = 1$? When $a+b = 2$? Interpret both with the transition diagram and with $\lambda$.
- $\lambda$ is the second eigenvalue of $P$ (the first is 1). Verify by computing the characteristic polynomial.
- Write the cut-balance equation $\pi_0 a = \pi_1 b$ and derive $\pi$ from it in one line — then compare with the limit above.

**AI study card** — paste one of these into your AI assistant:

- `Walk me through the two-state Markov chain P^n derivation by asking one question per step; I must supply the recursion and the fixed point myself.`
- `Give me three two-state chains (a+b<1, a+b=1, a+b>1) and make me predict, before computing, whether P^n converges monotonically, instantly, or with oscillation.`

## Proof 5 — Solving the stationary equations: the umbrella chain

**Theorem.** For the umbrella chain with $r \ge 1$ umbrellas and rain probability $p \in (0,1)$ (CP3 #5), the unique stationary distribution is

$$\pi_0 = \frac{1-p}{r+1-p}, \qquad \pi_1 = \cdots = \pi_r = \frac{1}{r+1-p},$$

and the long-run fraction of wet trips is $\varphi = p\,\pi_0 = \dfrac{p(1-p)}{r+1-p}$.

*Proof.* Transitions: $0 \to r$ w.p. 1; for $i \geq 1$, $i \to r-i$ w.p. $1-p$ and $i \to r-i+1$ w.p. $p$. Write the balance equation $\pi_j = \sum_i \pi_i P_{ij}$ for each *target* $j$ by asking "who leads to $j$?":

$$\begin{aligned} j = r: &\quad \pi_r = \pi_0 \cdot 1 + \pi_1 \cdot p \qquad \text{(from 0, and from 1 in rain)}, \\ 1 \leq j \leq r-1: &\quad \pi_j = \pi_{r-j}\,(1-p) + \pi_{r-j+1}\,p, \\ j = 0: &\quad \pi_0 = \pi_r \cdot (1-p) \qquad \text{(from } r \text{ with no rain)}. \end{aligned}$$

Try $\pi_1 = \cdots = \pi_r = c$. The middle equations become $c = c(1-p) + cp$, true. The last gives $\pi_0 = c(1-p)$, and then the first reads $c = c(1-p) + cp$ — true again. Normalising, $c(1-p) + r\,c = 1 \Rightarrow c = \dfrac{1}{r+1-p}$. The chain is finite and irreducible (every state leads to $r$, and $r$ leads to every state via $r-i$ and $r-i+1$), so this solution is the unique one. Finally she gets wet on a trip iff it rains *and* she is in state 0; rain is independent of the past, so $\varphi = P(\text{rain})\,P(X_n = 0) \to p\,\pi_0$. ☐

*Cost (CP5 #3).* $\tau(r) = c\varphi(r) + dr$ is convex in $r$; the continuous minimiser is $r^* = \sqrt{c\,p(1-p)/d} - (1-p)$, and the optimal integer is one of its neighbours.

**Checkpoint** — answer before moving on.

- Which single balance equation was redundant here? Show that adding all $r+1$ equations gives $0 = 0$.
- Why is $\varphi$ maximal at $p = \frac{1}{2}$ and not at $p \to 1$? Give the one-sentence reason from the formula and from the story.
- Verify aperiodicity: find the self-loop for even $r$ and for odd $r$.

**AI study card** — paste one of these into your AI assistant:

- `Ask me to set up (not solve) the balance equations for the umbrella chain with r=3; check each equation against "who leads into this state".`
- `Claim (falsely) that the fraction of wet trips is pi_0. Make me find the missing factor and explain why the two events are independent.`

## Proof 6 — Long-run fractions, mean return times, and the cost formula

**Theorem.** Let the chain be irreducible and positive recurrent on a finite $\mathcal{S}$, with $m_i$ the mean return time to $i$. Then, with probability 1,

$$\text{(a)}\ \frac{1}{n}\sum_{k=1}^{n} \mathbf{1}\{X_k = i\} \ \to\ \frac{1}{m_i} = \pi_i, \qquad \text{(b)}\ \frac{1}{n}\sum_{k=1}^{n} C(X_k)\ \to\ \sum_{i\in\mathcal{S}} C(i)\,\pi_i.$$

*Proof of (a) (time fraction $= 1/m_i$).* Let $T_1 < T_2 < \ldots$ be the successive visit times to $i$. By the Markov property and homogeneity, the gaps $T_{\ell+1} - T_\ell$ are i.i.d. with mean $m_i$ (each restarts the chain from $i$). Let $N(n)$ be the number of visits by time $n$; then $T_{N(n)} \leq n < T_{N(n)+1}$, so

$$\frac{T_{N(n)}}{N(n)} \ \leq\ \frac{n}{N(n)}\ <\ \frac{T_{N(n)+1}}{N(n)+1}\cdot\frac{N(n)+1}{N(n)}.$$

Since $i$ is recurrent, $N(n) \to \infty$, and by the strong law of large numbers both outer terms converge to $m_i$. Hence $n/N(n) \to m_i$, i.e. $N(n)/n \to 1/m_i$. That this limit is the solution $\pi_i$ of $\pi = \pi P$ is the content of the limiting theorem (§1.5), which we accept. ☐

*Proof of (b).* Group the terms by state:

$$\frac{1}{n}\sum_{k=1}^{n} C(X_k) = \sum_{i\in\mathcal{S}} C(i)\,\frac{1}{n}\sum_{k=1}^{n} \mathbf{1}\{X_k = i\} \;\to\; \sum_{i\in\mathcal{S}} C(i)\,\pi_i,$$

using (a) for each of the *finitely many* states (the limit of a finite sum is the sum of the limits). □

*Consequences.* Mean return time $m_i = 1/\pi_i$ (CP5 #1: a down period recurs every $1/\pi_{\text{down}} \approx 49.5$ days; CP5 #4: mean tool lifetime $= 1/\pi_0$). Time averages (Kim's own spending) and ensemble averages (revenue per student) coincide — *ergodicity* (CP5 #5).

**Checkpoint** — answer before moving on.

- In (a), where exactly is positive recurrence (finite $m_i$) needed for the sandwich argument to give a finite, nonzero limit?
- CP5 #4: mean lifetime $= 1/\pi_0$ and also $= \sum_{k\ge 0} P(L > k)$ (tail sum). Explain why both are the same number without computing either.
- Part (b) exchanged a limit with a finite sum. Where could it fail for an infinite state space with unbounded $C$?
- **(midterm-style)** With the 3-state $P$ of Proof 2's last checkpoint, let $X_n$ be the number of movies rented on the weekly trip $n$, at 1,500 won per movie plus a 500 won bus fare. Find $\pi$ and the long-run average weekly cost. (Answer: $\pi = (\frac{1}{3}, \frac{1}{9}, \frac{5}{9})$, cost $\approx$ 2,333 won.)

**AI study card** — paste one of these into your AI assistant:

- `Quiz me on why the long-run fraction of time in state i equals 1/m_i. Force me to write the sandwich inequality with the visit times and to say where the SLLN is used.`
- `Give me a 3-state chain with costs; I compute the long-run average cost by hand and you check it. Then ask me whether the answer changes with the initial state, and why.`

## Proof 7 — PageRank: taxation makes the surfer ergodic (exam topic)

**Theorem.** Let $P$ be an $n \times n$ row-stochastic matrix (dead ends replaced by uniform rows) and $0 < \beta < 1$. Then $G = \beta P + \frac{1-\beta}{n}\mathbf{1}\mathbf{1}^\top$ is row-stochastic and strictly positive, hence ergodic, so $\pi = \pi G$, $\pi\mathbf{1} = 1$ has a unique solution. Moreover, for any probability row vector $p_0$ and $p_{k+1} = p_k G$,

$$\|p_{k+1} - \pi\|_1 \;\le\; \beta\,\|p_k - \pi\|_1, \qquad \text{so} \qquad \|p_k - \pi\|_1 \le \beta^{\,k}\|p_0 - \pi\|_1 \to 0.$$

*Proof.* Row sums: $\sum_j G_{ij} = \beta\cdot 1 + (1-\beta)\cdot\frac{n}{n} = 1$, and every entry is at least $\frac{1-\beta}{n} > 0$. A positive matrix makes every state accessible from every other in one step (irreducible) and gives every state a self-loop (aperiodic); a finite irreducible chain is positive recurrent (Proof 3(c)). So the chain is ergodic and $\pi$ is unique.

For the rate, let $e_k = p_k - \pi$. Both $p_k$ and $\pi$ sum to 1, so $e_k\mathbf{1} = 0$. Then, using $\pi = \pi G$,

$$e_{k+1} = p_k G - \pi G = e_k G = \beta\, e_k P + \frac{1-\beta}{n}\,(e_k\mathbf{1})\mathbf{1}^\top = \beta\, e_k P.$$

Finally $\|e_k P\|_1 = \sum_j \left|\sum_i (e_k)_i P_{ij}\right| \le \sum_i |(e_k)_i| \sum_j P_{ij} = \|e_k\|_1$ because $P$ is row-stochastic. Hence $\|e_{k+1}\|_1 \le \beta \|e_k\|_1$. □

*Reading it.* The web's structure never entered the rate: with $\beta = 0.85$, about 50 iterations give $10^{-4}$ accuracy on *any* web (practicum §5). A page with no incoming links has exactly $\pi_j = (1-\beta)/n$ (practicum §6). Without taxation ($\beta = 1$) the bound is useless, and spider traps/dead ends break uniqueness (practicum §3). *(verified in notebook: PageRank practicum: $\pi = (0.4, 0.4, 0.2)$; with tax $(0.26, 0.19, 0.55)$; $|\lambda_2| \le \beta$)*

**Checkpoint** — answer before moving on.

- Prove $\pi_j = (1-\beta)/n$ for a page with no incoming links directly from $\pi = \pi G$.
- Why does a dead end have to be repaired *before* taxation? What goes wrong with the row sums otherwise?
- The 3-cycle (period 3) has no limit; after taxation it does. Which line of the proof kills the period?
- **(midterm-style)** Sites A, B, C: A links to A, B, C; B has no outbound links; C links to A and B. Assign the transition probabilities (repair the dead end), tax with $\beta = 0.85$, and rank. Two sites tie — say which and why before computing (practicum §6, Drill B).

> **AI study card** — paste one of these into your AI assistant:
>
> - `Quiz me step by step: show that the taxed PageRank matrix is ergodic and that power iteration contracts the L1 error by the factor beta. Make me justify e_k 1 = 0.`
> - `Give me a 4-page web with one spider trap. I must (i) compute pi without tax, (ii) explain what is wrong, (iii) compute pi with beta=0.8 by hand for one iteration. Check every number.`

## Proof 8 — Subsampling a Markov chain (midterm-style)

**Theorem.** Let $\{X_n\}$ be a time-homogeneous DTMC with matrix $P$ and fix $k \ge 1$. Then $Y_n = X_{kn}$ ($n \ge 0$) is a time-homogeneous DTMC with transition matrix $P^k$. In particular $X_0, X_2, X_4, \ldots$ is a Markov chain.

*Proof. Step 1 (the Markov property reaches r steps ahead and ignores any past event).* Let $A$ be any event determined by $X_0, \ldots, X_{m-1}$ with $P(X_m = i, A) > 0$. Summing over the intermediate path $X_{m+1} = j_1, \ldots, X_{m+r-1} = j_{r-1}$ and applying the one-step Markov property together with time-homogeneity at each step,

$$P(X_{m+r} = j \mid X_m = i, A) = \sum_{j_1, \ldots, j_{r-1}} P_{ij_1} P_{j_1 j_2} \cdots P_{j_{r-1} j} = P^{(r)}_{ij} = (P^r)_{ij},$$

which does not depend on $A$ (nor on $m$). *Step 2.* Take $m = kn$, $r = k$ and $A = \{X_{k(n-1)} = i_{n-1}, \ldots, X_0 = i_0\}$, which is determined by $X_0, \ldots, X_{kn-1}$:

$$P(Y_{n+1} = j \mid Y_n = i, Y_{n-1} = i_{n-1}, \ldots, Y_0 = i_0) = P(X_{k(n+1)} = j \mid X_{kn} = i, A) = (P^k)_{ij}.$$

The right-hand side depends only on $(i, j)$: this is the Markov property for $\{Y_n\}$, and the transition matrix is the same for every $n$, so $\{Y_n\}$ is time-homogeneous with matrix $P^k$ (a stochastic matrix, since products of stochastic matrices are stochastic). □

**Checkpoint** — answer before moving on.

- Where exactly was time-homogeneity used, and what would change without it (write the matrix of $\{Y_n\}$ at step $n$)?
- Is $X_0, X_1, X_3, X_6, X_{10}, \ldots$ (gaps $1, 2, 3, 4, \ldots$) a Markov chain? Is it time-homogeneous?
- Subsample the 3-cycle (period 3) with $k = 3$: what is $P^3$, and what does the result say about periodic chains and subsampling? Is $\{Y_n\}$ still irreducible?

> **AI study card** — paste one of these into your AI assistant:
>
> - `Quiz me: prove that X_0, X_2, X_4, ... is a Markov chain. Make me define "the past" of the subsampled chain as an event about the original chain and check that the Markov property handles it.`
> - `Give me a 3-state P and ask me for the transition matrix of the chain observed every second step. Then ask whether its stationary distribution changes. Check my answers.`

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**Where to next.** CP5 wrap-up → these proofs (gap-fill first) → Checkpoint quiz at the start of the next lecture → HW 2 Part A (closed-book style, no AI on the first attempt) → PageRank practicum §6 drill.